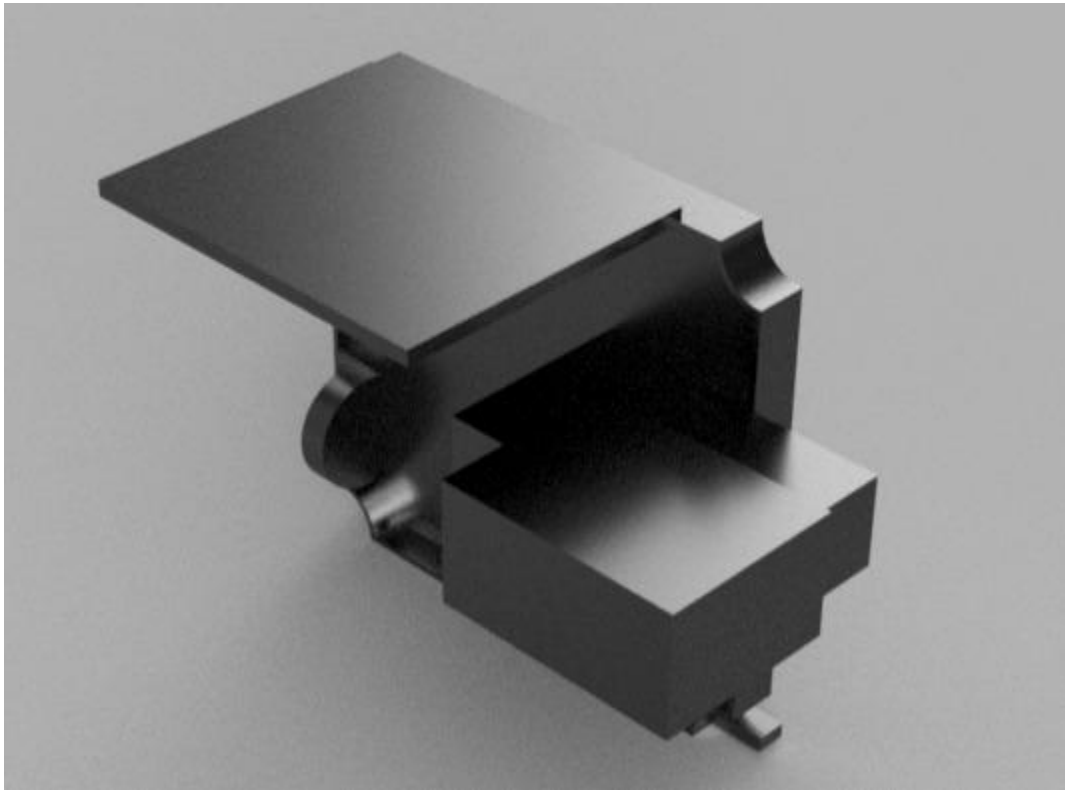


PROJECT: Premium Mini-ITX Aluminum PC Case

HERO RENDER:



Design Overview:

Internal Layout:

This analysis demonstrates the design's internal planning and structural support.

- **Primary Structure:**
Outer Shell & Motherboard Tray.
- **Component Housing:**
Dedicated mounting points for PSU, GPU, and cooling fans.
- **Cable Routing:**
Integrated management channels for optimized airflow.

DESIGN GOALS:

Thermal Optimization:

Engineered for high-efficiency airflow to ensure consistent cooling for high-performance components, minimizing thermal throttling under heavy loads.

Structural Minimalism:

Achieving a balance between structural rigidity and lightweight design through the strategic use of material thickness and geometry.

Modular Component Support:

Designed to accommodate diverse hardware configurations, ensuring universal compatibility for motherboards, GPUs, and various storage drives.

Integrated Cable Management:

Features dedicated, hidden routing channels to sequester cabling, thereby reducing airflow obstruction and enhancing visual cleanliness.

User-Centric Accessibility:

Focused on intuitive assembly and maintenance, providing tool-less or easy-access panels for upgrades.

Acoustic Dampening:

Incorporating design elements that reduce vibration and fan noise to ensure a quiet operation even under intensive computing.

Future-Proof Expansion:

Proactive space allocation for future hardware standards, including larger cooling solutions and liquid cooling radiators.

Form Factor Efficiency:

Maximizing internal usable volume within a compact exterior footprint, ideal for modern, clutter-free workspaces.

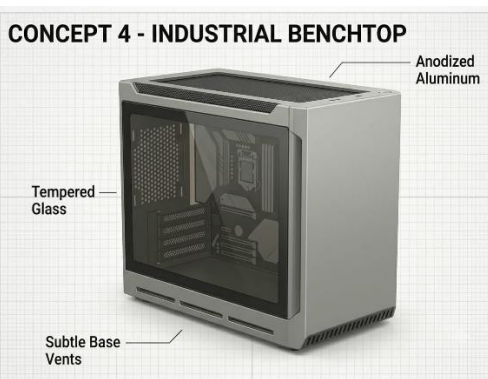
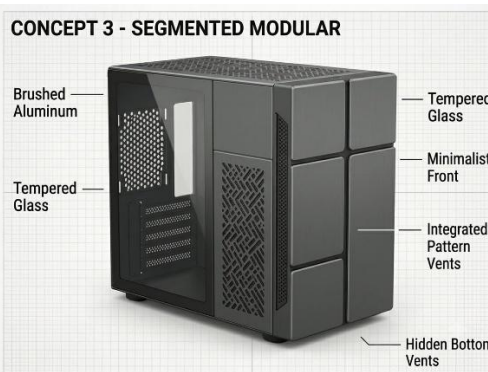
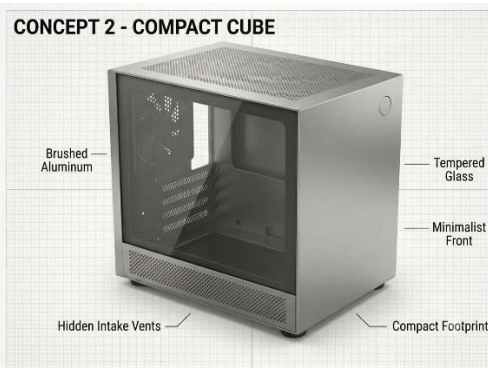
Environmental Durability:

Using materials and finishing techniques that ensure resistance to wear, dust accumulation, and daily operational heat.

Aesthetic Synergy:

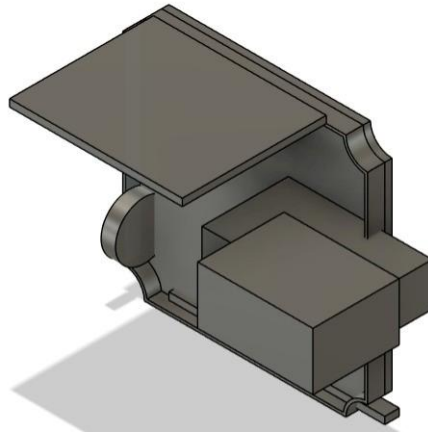
Adopting a sleek, professional design language that complements both high-end gaming setups and minimalist office environments.

CONCEPT SKETCHES:

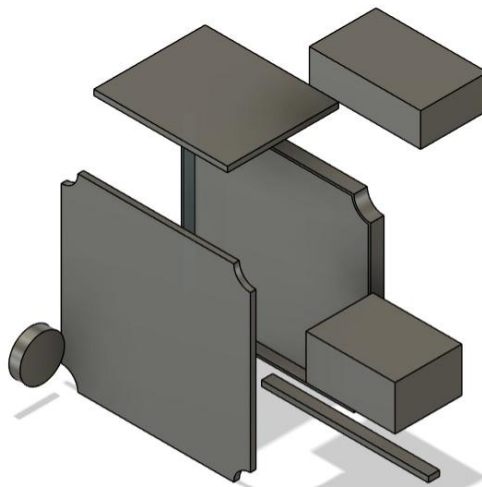


CAD MODEL SCREENSHOTS:

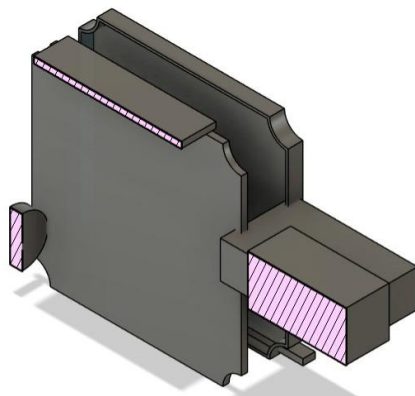
Assembly view:



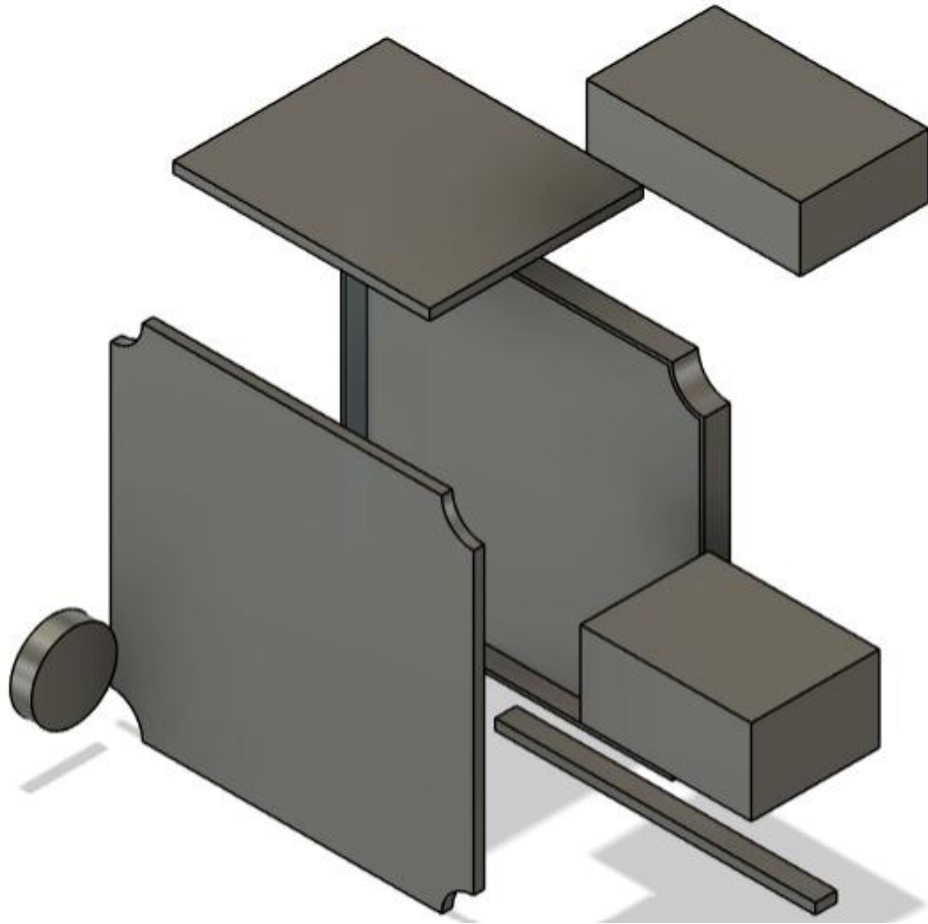
Exploded view:



Cross section:



EXPLODED VIEW:



Exploded Assembly Analysis:

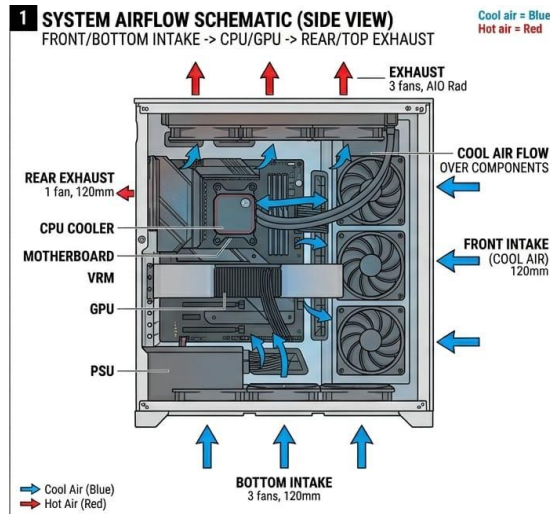
This view illustrates the modular assembly of the Apex One PC Case, verifying structural fitment and component hierarchy.

Assembly Logic (DFA): Demonstrates the efficient 'Design for Assembly' sequence, highlighting how individual modules fit together to form a cohesive, rigid structure.

Modular Maintenance: Showcases the modularity of the design, ensuring each component can be easily accessed or replaced without disassembling the entire chassis.

Spatial Precision: Visualizes the precise clearance and fitment between internal mounts and the outer shell, confirming a clutter-free integration of high-performance hardware.

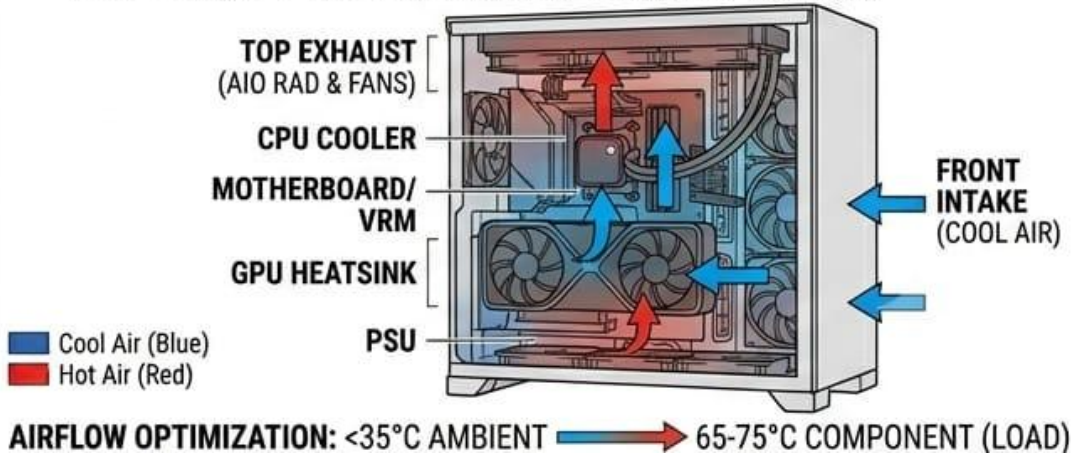
AIRFLOW ANALYSIS:



3 INTERNAL COMPONENT INTERACTION

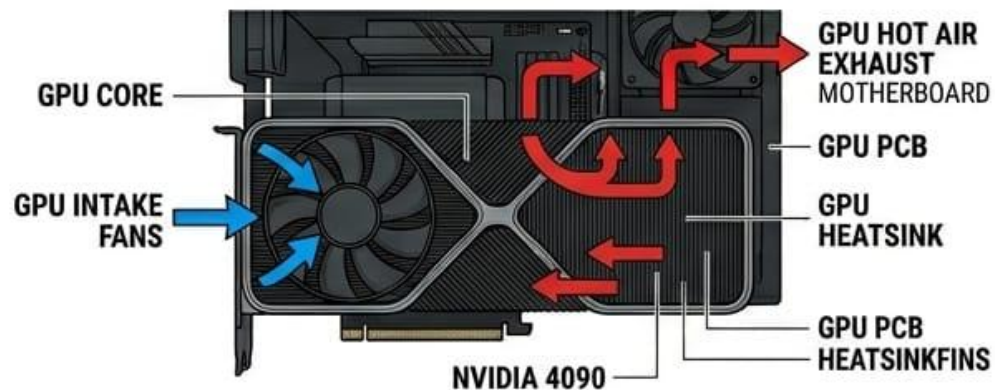
FRONT INTAKE -> GPU -> CPU COOLER -> REAR/TOP EXHAUST

Cool Air (Blue)
Hot Air (Red)



2 DEDICATED GPU COOLING PATH (TOP VIEW)

ISOLATED GPU CHAMBER INTAKE & DIRECT HEATSINK EXHAUST



MANUFACTURING DRAWINGS:

SHEET METAL BEND DRAWINGS:

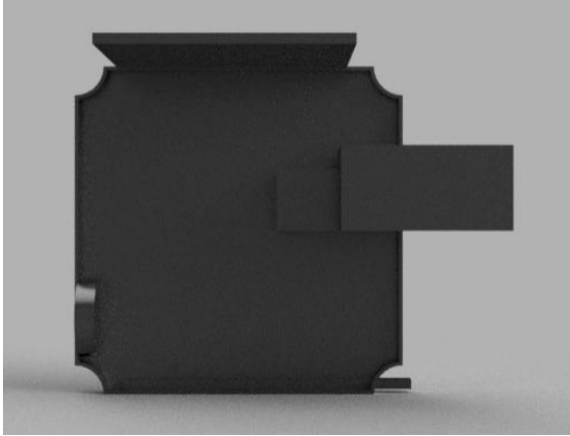
BEND NO.	BEND ANGLE	BEND RADIUS	LOCATION
Bend #1	90°	2.0 mm	Front Edge
Bend #2	90°	2.0 mm	Rear Edge
Bend #3	90°	3.0 mm	Top Flange
Bend #4	45°	2.5 mm	Bottom Rail

DIMENSIONSED TECHNICAL DRAWINGS:

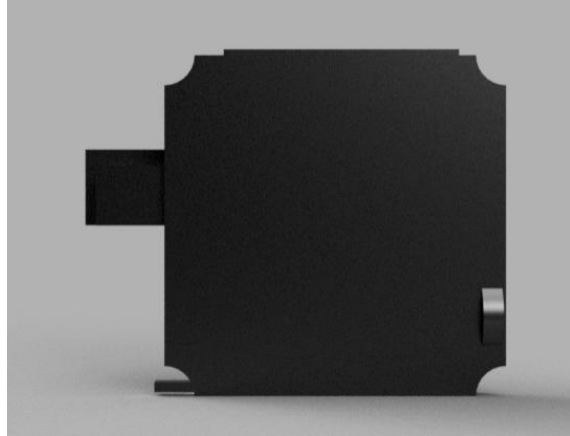
COMPONENT	LENGTH (mm)	WIDTH (mm)	HEIGHT (mm)
Outer Shell (Main Body)	170 mm	170 mm	2 mm
Front Panel	200 mm	15 mm	420 mm
Side Panel (Glass)	420 mm	4 mm	380 mm
Motherboard Tray	100 mm	120 mm	5 mm
PSU Mount Area	80 mm	60 mm	40 mm
GPU Mount Area	100 mm	30 mm	60 mm
Fan Mount	20 mm	20 mm	10 mm
Cable Channel	150 mm	10 mm	5 mm

FINAL RENDERS:

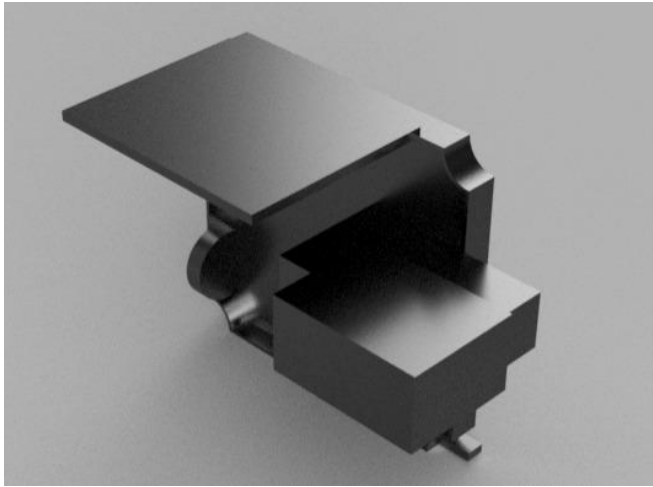
FRONT VIEW:



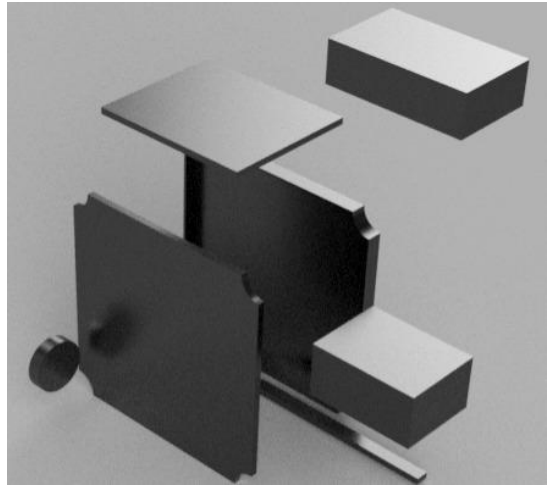
REAR VIEW:



PERSPECTIVE VIEW:



EXPLODED ASSEMBLY VIEW:



DESK SETUP RENDER:

